



ME 300: Engineering Thermodynamics

Example: Gibbs Equation of State

Example

A rigid tank with 0.3 kg of nitrogen (N_2) begins at a pressure of 506625 Pa and a temperature of 400 K. Heat is removed until the temperature is equal to 300 K. Calculate the change in entropy of the nitrogen. Properties of N_2 : $MW=28$ kg/kmol, $c_p=1040$ J/kg-K, $c_v=743$ J/kg-K.

Given: 0.3 kg
 $P_1 = 506625$ Pa
 $T_1 = 400$ K, $T_2 = 300$ K

$$\Delta S = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right) \sim \left[\frac{J}{kg-K} \right]$$
$$\Rightarrow \Delta S = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{P_2}{P_1} \right)$$

Find: $\Delta S \sim \left[\frac{J}{K} \right]$

$$\Delta S = c_v \ln \left(\frac{T_2}{T_1} \right) = 743 \ln \left(\frac{300}{400} \right) = -213.7 \frac{J}{kg-K}$$

Assume: ideal gas
 $c_v, c_p = \text{const}$
quasi-st.
 $\Delta T = 0$, $M = \text{const}$
 $\Delta U = 0$

$$\Delta S' = M \Delta S$$

$$\Delta S' = -64.12 \frac{J}{K}$$

$$P_2 = \frac{RT_2}{v_2} \quad v_2 = v_1$$
$$v_1 = \frac{RT_1}{P_1}$$
$$R = R_u / MW$$